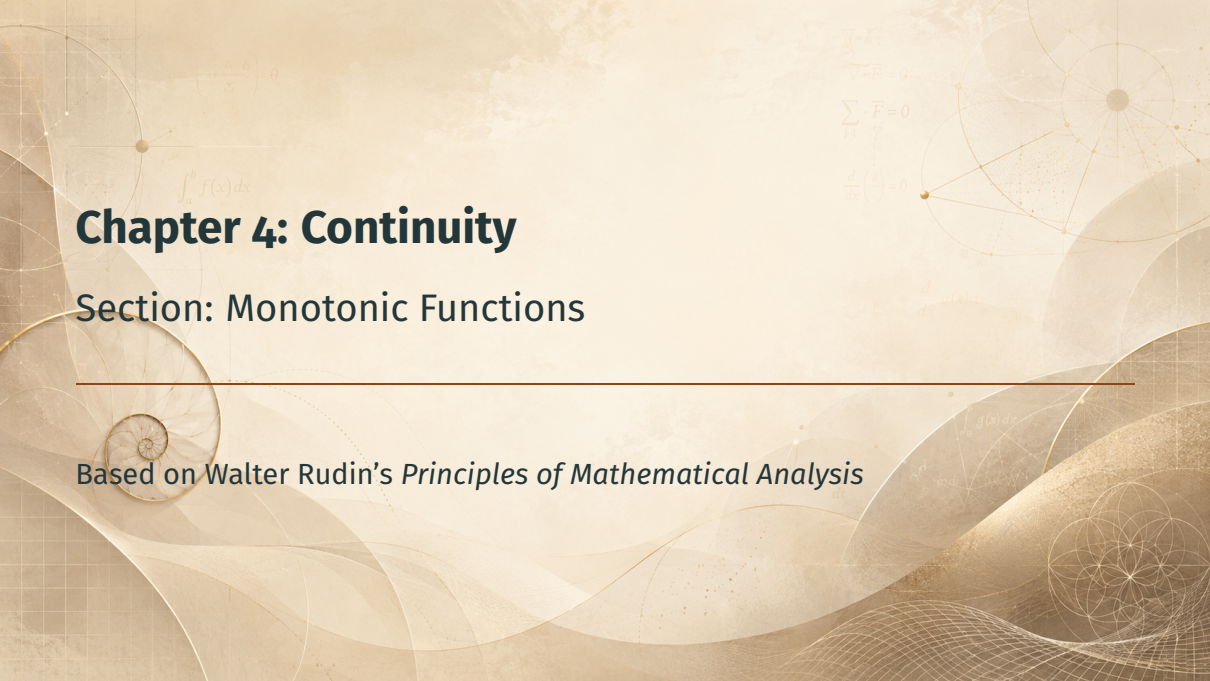


The background is a light beige color with various mathematical and geometric elements. In the top left, there is a formula $\left(\frac{a_1}{b_1}, \frac{a_2}{b_2} \right) = \theta$. Below it is $\int_a^b f(x) dx$. In the top right, there are formulas $\vec{\nabla} \cdot \vec{F} = 0$, $\sum_{i=1}^n \vec{F}_i = 0$, and $\frac{d}{dx} \left(\frac{g}{f} \right) = 0$. On the right side, there is a diagram of a sphere with lines connecting points on its surface. In the bottom right, there is a diagram of a circle with internal lines forming a star-like pattern. A horizontal line is drawn across the middle of the page.

Chapter 4: Continuity

Section: Monotonic Functions

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

Definition

One-Sided Limits

Countably Many Discontinuities

Discontinuities Can Be Dense

Summary

Motivation

Why Monotonic Functions?

- Functions that **never decrease** (or never increase) on an interval

$$\int_a^b f(x) dx$$

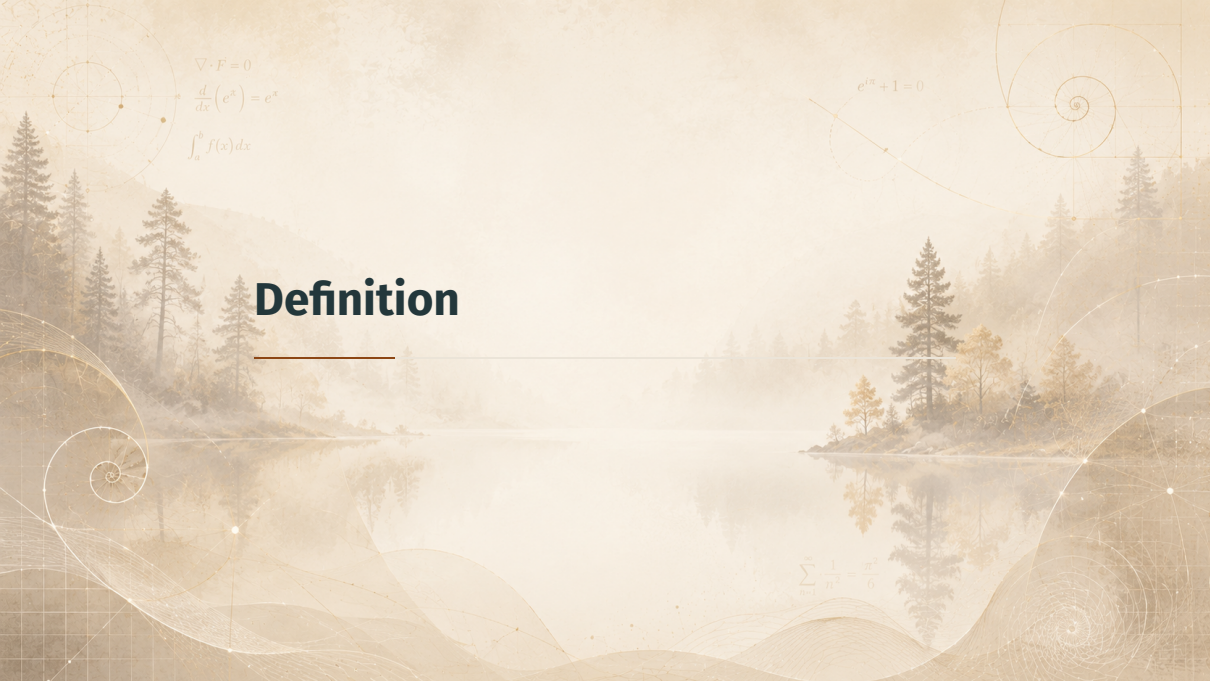
Why Monotonic Functions?

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- One-sided limits **always exist** — no wild oscillation possible

Why Monotonic Functions?

- Functions that **never decrease** (or never increase) on an interval
- One-sided limits **always exist** — no wild oscillation possible
- Discontinuities are **at most countable** — and can be dense!

$$\int_a^b f(x) dx$$


$$\nabla \cdot F = 0$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\int_a^b f(x) dx$$

$$e^{i\pi} + 1 = 0$$

Definition

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Definition 4.28: Monotonic Functions

Definition 4.28

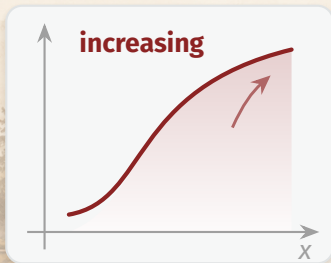
Let f be real on (a, b) .

- f is **monotonically increasing** if $a < x < y < b \Rightarrow f(x) \leq f(y)$.
- f is **monotonically decreasing** if $a < x < y < b \Rightarrow f(x) \geq f(y)$.

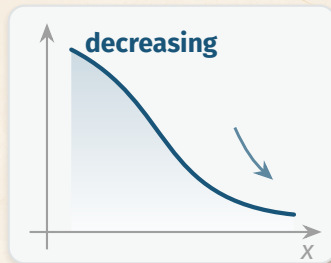
The *monotonic* functions are the increasing ones together with the decreasing ones.

Two Pictures

$$\frac{d}{dx}(e^x) = e^x$$
$$\int_a^b f(x) dx$$



$$x < y \Rightarrow f(x) \leq f(y)$$



$$x < y \Rightarrow f(x) \geq f(y)$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$e^{i\pi} + 1 = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

One-Sided Limits

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Theorem 4.29: One-Sided Limits Exist

Theorem 4.29

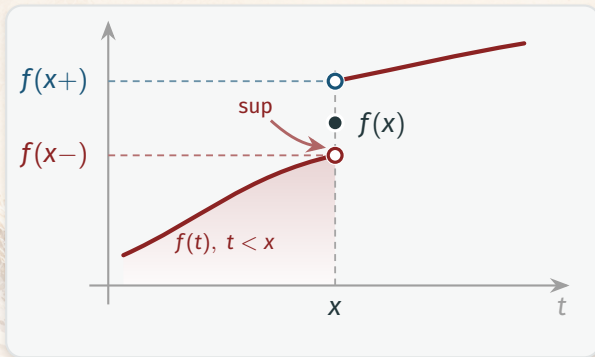
Let f be monotonically increasing on (a, b) . Then $f(x+)$ and $f(x-)$ exist at every $x \in (a, b)$:

$$\sup_{a < t < x} f(t) = f(x-) \leq f(x) \leq f(x+) = \inf_{x < t < b} f(t). \quad (25)$$

Furthermore, if $a < x < y < b$, then

$$f(x+) \leq f(y-). \quad (26)$$

Picture: The Left-Hand Limit as a Supremum



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$y = \frac{\sin x}{x}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Proof of 4.29 — Step 1: Name the Supremum

Step 1: Set up

Values $f(t)$ for $a < t < x$ are bounded above by $f(x)$. Let

$$A = \sup_{a < t < x} f(t), \quad A \leq f(x).$$

Goal: show $A = f(x-)$.

Proof of 4.29 — Step 2: Use the Supremum

Step 2: Approximate from below

Given $\varepsilon > 0$, by definition of A there is $\delta > 0$ with $a < x - \delta < x$ and

$$A - \varepsilon < f(x - \delta) \leq A. \quad (27)$$

Proof of 4.29 — Step 3: Squeeze with Monotonicity

Step 3: Trap all nearby values

For $x - \delta < t < x$, monotonicity gives

$$f(x - \delta) \leq f(t) \leq A. \quad (28)$$

Combining (27) and (28): $|f(t) - A| < \varepsilon$ for $x - \delta < t < x$.

$$f(x-) = A = \sup_{a < t < x} f(t)$$

Proof of 4.29 — Step 4: Comparing Across Points

Step 4: Why $f(x+) \leq f(y-)$

For $a < x < y < b$:

$$f(x+) = \inf_{x < t < b} f(t) = \inf_{x < t < y} f(t), \quad (29)$$

$$f(y-) = \sup_{a < t < y} f(t) = \sup_{x < t < y} f(t). \quad (30)$$

Every value on (x, y) lies between these $\Rightarrow f(x+) \leq f(y-)$.

Corollary: No Second-Kind Discontinuities

Corollary

Monotonic functions have **no discontinuities of the second kind**.

Recall

A discontinuity of the second kind is one where a one-sided limit *fails to exist*. Theorem 4.29 guarantees both always exist.

The background is a complex, artistic composition. It features a landscape with mountains and wavy, ethereal lines in shades of blue and white. Overlaid on this are various mathematical and geometric elements: a golden spiral on the left, a grid of dots with connecting lines, and several mathematical formulas. The overall color palette is warm, with golden yellows and browns, contrasting with the cooler blues of the landscape.

Countably Many Discontinuities

$$\frac{d^2 y}{dx^2} + \omega^2 y = 0$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\nabla^2 \phi = 0$$

$$e^{im} + 1 = 0$$

$$\frac{\partial u}{\partial t} = \sigma \nabla^2 u$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Theorem 4.30: Discontinuities Are Countable

Theorem 4.30

Let f be monotonic on (a, b) . Then the set of points of (a, b) at which f is discontinuous is **at most countable**.

Strategy: attach a distinct rational number to each jump.

Proof of 4.30 — Step 1: Insert a Rational in Each Jump

Step 1: Define $r(x)$

Take f increasing; let E be its set of discontinuities. At each $x \in E$ we have $f(x-) < f(x+)$, so choose a rational $r(x)$ with

$$f(x-) < r(x) < f(x+).$$

Proof of 4.30 — Step 2: The Map Is Injective

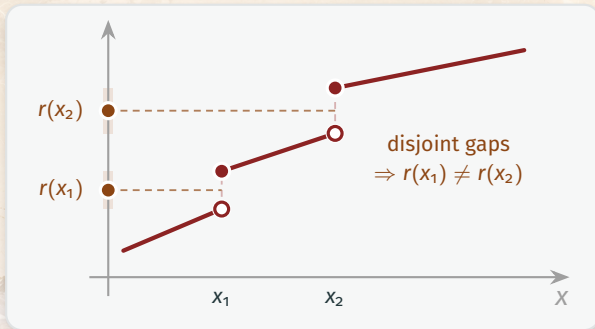
Step 2: Distinct points get distinct rationals

If $x_1 < x_2$, then by (26): $f(x_1+) \leq f(x_2-)$. Hence the jump intervals are disjoint and

$$r(x_1) < f(x_1+) \leq f(x_2-) < r(x_2) \Rightarrow r(x_1) \neq r(x_2).$$

$E \hookrightarrow \mathbb{Q}$, so E is at most countable.

A Picture of Disjoint Jumps



The background is a composite image. It features a misty, golden-hour landscape with a lake, mountains, and cherry blossom trees. Overlaid on this are various mathematical diagrams: a coordinate plane with a point labeled 'a', a complex plane with points 'z' and 'z-bar', a geometric diagram with a point 'P' and lines, a golden spiral, and several geometric constructions involving circles and triangles. The overall color palette is warm, with soft pinks, oranges, and yellows.
$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1 + \sqrt{5}}{2}$$

Discontinuities Can Be Dense

$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

Remark 4.31: Jumps Need Not Be Isolated

Remark 4.31

Given *any* countable set $E \subset (a, b)$ — even a **dense** one — there is a monotonic f discontinuous exactly at the points of E .

$$\nabla^2 \psi + k^2 \psi = 0$$

“At most countable” is sharp: countably many jumps can fill the line densely.

Remark 4.31 — The Construction

Build f from a convergent series

Enumerate $E = \{x_n\}$. Pick $c_n > 0$ with $\sum c_n$ convergent. Define

$$f(x) = \sum_{x_n < x} c_n \quad (a < x < b). \quad (31)$$

Sum over all n with $x_n < x$ (empty sum = 0).

Remark 4.31 — Properties of f

The function f from (31) satisfies

- (a) f is monotonically increasing on (a, b) ;
- (b) f is discontinuous at each x_n , with jump $f(x_n+) - f(x_n-) = c_n$;
- (c) f is continuous at every other point.

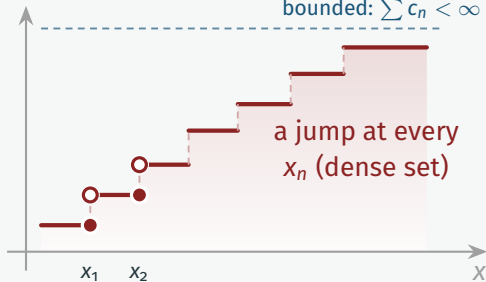
Also $f(x-) = f(x)$ everywhere: f is **continuous from the left**.

Picture: A Dense Set of Tiny Jumps

$$e^{i\pi} + 1 = 0$$

$$\phi = \frac{1+\sqrt{5}}{2}$$

bounded: $\sum c_n < \infty$



$$\nabla^2 \psi + k^2 \psi = 0$$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Summary

$$\varphi = \frac{1+\sqrt{5}}{2}$$

$$\nabla^2 \phi = 0$$

$$\frac{d}{dx} \sin x = \cos x$$

Summary

Key takeaways:

- **Def 4.28:** monotonic = never decreasing or never increasing

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$\varphi = \frac{1+\sqrt{5}}{2} \quad \nabla^2 \phi = 0$$

$$\frac{d}{dx} \sin x = \cos x$$

Summary

Key takeaways:

- **Def 4.28:** monotonic = never decreasing or never increasing
- **Thm 4.29:** both one-sided limits exist (sup / inf); no second-kind discontinuities

$$\frac{d}{dx} \sin x = \cos x$$

Summary

Key takeaways:

- **Def 4.28:** monotonic = never decreasing or never increasing
- **Thm 4.29:** both one-sided limits exist (sup / inf); no second-kind discontinuities
- **Thm 4.30:** discontinuities are at most countable — but (4.31) can be dense

Coming up next: Infinite Limits and Limits at Infinity.

$$\frac{d}{dx} \sin x = \cos x$$